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PAPER

Preparation and structure investigation of polyethylene terephthalate polyester reinforced $\text{Ni}_{0.5}\text{Zn}_{0.5}\text{Fe}_2\text{O}_4$ nanoparticles for gamma ray shielding performance

Meshari Almeshari¹, Fawzy Hammad Sallam^{2,*} , Mohamed Tharwat³, Yasser Alzamil¹, Mohammed Salih¹, Bader Alshoumr⁴, Amjad Alyahyawi¹ and Atef El-Taher⁵

¹ Department of Diagnostic Radiology, College of Applied Medical Sciences, University of Hail, Hail, Saudi Arabia

² Nuclear Materials Authority, Department of Geochemical Exploration, Cairo, Egypt

³ Physics and Engineering Mathematics Department, Faculty of Electronic Engineering, Menoufia University, Menouf 32952, Egypt

⁴ Department of Health informatics, College of Public Health and Health informatics, University of Hail, Hail, Saudi Arabia

⁵ Physics Department, Faculty of Science, Al-Azhar University, Assuit, Egypt

* Author to whom any correspondence should be addressed.

E-mail: m.almeshari@uoh.edu.sa, Fozsallam2010@yahoo.com, Mohamed.attia@el-eng.menofia.edu.e.g, Y.alzamil@uoh.edu.sa, Mo.salih@uoh.edu.sa, B.alshoumr@uoh.edu.sa, A.alyahyawi@uoh.edu.sa and atef.eltaher@gmail.com

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Abstract

Defects of high atomic materials gamma-ray shielding such as low chemical stability, low mechanical properties, and heaviness lead us to investigate other light and flexible materials such as polymers. Polymer-doped nanosized materials are the most frequently examined materials. In this study, polyethylene terephthalate $[(\text{C}_{10}\text{H}_8\text{O}_4)_n]$ was doped with $\text{Ni}_{0.5}\text{Zn}_{0.5}\text{Fe}_2\text{O}_4$ nanoparticles up to 40 wt% (0.0, 10.0, 20.0, 40.0 wt%) prepared by Sol–Gel auto-combustion method with the help of Gelatin. The polyester/Nanofiller composite structures were identified using x-ray photoelectron spectroscopy, x-ray diffraction, Scanning, and Transmission electron microscope as well as density measurements. x-ray photoelectron spectroscopy confirmed the successful doping of nanofiller in the polyester structure as Zn signals appear in the atomic composition and Fe signals appear in the deconvolution of the peaks. x-ray diffraction, transmission, and scanning electron microscope display the same result. x-ray diffraction graph information with the Scherer equation offered the crystal size of the composite (26 nm). Polyester/nanofiller samples were scanned against gamma-ray and experimental shielding factors were computed using a narrow beam transmission technique with sodium iodide detector and two-point sources Cs-137 and Co-60. Experimental Linear and mass attenuation coefficient values swelled as percentages of nanofiller increased in the polyester structure. Experimental Mass attenuation values were compared with theoretical ones estimated from XCOM and Physics-X programs. The difference between them does not exceed 12% which is acceptable as the x-ray photoelectron spectroscopy atomic composition utilized in the theoretical data calculation does not reveal Ni signals. This may occur at the depth of the composite structure. Finally, the half-value layer, the Tenth value layer, and the Mean free path are determined experimentally, and their values are reduced as the nanofiller doping percentage rises in the structure. This result confirms the efficiency of nanofiller addition to the polyester structure to attenuate gamma-ray.

1. Introduction

Human exposure to high electromagnetic radiation, such as gamma-ray and x-ray can be deleterious to the health because extended exposure causes radiation dosage accumulation which leads to several health risks. These risks include cell mutations, organ failure, carcinogenesis, and other symptoms [1–5]. Various shielding materials have been investigated to absorb or attenuate hazardous radiation, especially gamma-ray such as Pb-based compounds and heavy metals [6, 7].

Shielding gamma-ray requires two specified conditions: high density and high Z number (atomic number); therefore low-Z materials such as polymers alone do not achieve the required attenuation. Many researchers have focused on constructing compounds from polymers and other high-Z and high-density materials, such as by the addition Pb to the polymer structure. Pb is considered the superior shield of gamma-ray as it has high Z also high density [8–12].

Polymers containing nano- or micro-compounds such as metal oxide, metals, or clay particles have been the most investigated materials against gamma-ray in recent years [13, 14]. These materials are incorporated between nano- or micro-fillers and the polymer matrix [15]. These nanofillers/polymer matrices aim to enhance the gamma-ray shielding characteristics. Doping is a lightweight and flexible material that can protect from gamma-ray [16, 17]. Most of the previous studies focus on the relation between filler size and how it affects the lattice also the characteristics of gamma-ray shielding. Nano-size filler seems to be the best material to accomplish this job.

Organic polymers are considered major hosts for trapping semiconductor nanoparticles and metals. Nanoparticle materials have excellent properties and larger surface area than conventional materials so it is better to add to polymer nanomaterials for superior properties (polymer/nanofiller composites). Polyester is the fourth most produced polymeric material, accounting for 18% of the polymer industry [18–23]. Lately, ferromagnetic particles combined with polymer composites are utilized as microwave absorbents at high frequencies [24–26]. NiZn ferrite showed excellent mechanical strength, notable chemical stability, and saturation magnetization in addition to Encouraging results for electromagnetic wave attenuation this is due to natural resonance. NiZn ferrite nanomaterials are used as an EMI shielding material [27, 28]. Therefore, ferromagnetic particles combined with polymer composites are effective, lightweight, and flexible for gamma-ray shielding applications and other applications like radio-absorbing materials, microwave Inductors, and microwave transmitters [29–34].

Mahmoud *et al* [35], investigated high-density polyethylene (HDPE) loaded with lead oxide nanoparticles (PbO-NPs) up to 50.0 wt% and concluded that better shielding characteristics are achieved when nanoparticles increase in the polymer structure.

Natural rubber is loaded with lead as filler up to 100 phr and the matrix is investigated as a gamma-ray shield. The matrix showed remarkable shielding properties, especially at higher Pb content [36]. High-density polyethylene is filled with different concentrations of WO₃ nanoparticles (two different sizes of nanoparticles). The composite structure was investigated and found that WO₃ nanoparticles have a uniform dispersion at low-weight fractions. The composite was tested against x-rays and showed that shielding parameters increased with WO₃ nanoparticles percentage [37].

Polyvinyl chloride (PVC) reinforced chalcocite and hematite minerals were investigated by simulation against gamma-ray and found that both minerals increased shielding parameters [38]. Siloxane polymers filled with boron element composite are investigated and showed good radiation shielding ability for both neutrons and gamma-ray [39]. Poly methyl methacrylate (PMMA) polymer-reinforced micro and nanoparticles of ZrO₂ as a matrix are prepared and measured as a gamma-ray shield. The results showed that increasing micro and nanoparticles of ZrO₂ in the polymer composite increases μ values while the half-value layer and tenth value layer decrease. Nano filler increased the linear attenuation coefficient by 17.84% more than microparticles at the same weight percent and the same energy [40].

For the above investigations, the choice of the polyester matrix (polyethylene terephthalate (PET)) is due to its good properties such as being colorless, lightweight, thermoplastics, robust, semi-rigid to rigid, mechanically resistant to impact, stretchable during processing, extremely inert, shows gas barrier against moisture and CO₂, can blended with other polymers and it can be copolymerized. PET has more sustainability than other polymers. PET is involved in 16% of plastic usage in the European packaging industry also it monopolizes the bottle fabrication industry [41].

PET showed the highest linear attenuation coefficient than other commercial polymers also HVL values are close to ordinary concrete and commercial glass, especially at low energy. Polymer materials are distinguished in gamma-ray shielding as it has low fabricating cost, durability, the capability to create complex forms, and optical transparency. For the above reasons, polymers are preferred over traditional materials (Lead) that are expensive, heavyweight, and toxic. The properties of such polymer materials can be enlarged by doping with nano-fillers [42].

This work aims to prepare Ni_{0.5}Zn_{0.5}Fe₂O₄ nanocomposites and doping them to polyethylene terephthalate up to 40% doping also characterizing the structure of lightweight polyester doped with a high concentration of Ni_{0.5}Zn_{0.5}Fe₂O₄ nanocomposites in addition to testing and calculating the shielding properties of that composite against gamma-ray.

2. Materials and method

2.1. Composite preparation

$\text{Ni}_{0.5}\text{Zn}_{0.5}\text{Fe}_2\text{O}_4$ nanocomposites were prepared successfully using sol-gel auto-combustion method with the help of gelatin in which metal nitrates $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$, $(\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O})$ and $\text{Ni}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ are weighted according to formula $\text{Zn}_x\text{Ni}_{1-x}\text{Fe}_2\text{O}_4$ where $x = 0.5$. 5 g of commercial gelatin was dissolved in hot distilled water in addition to the weighted powder also dissolved in distilled water. The solution was held under a mechanical stirrer with a stirring speed of 120 min and an almost steady temperature reached 353 K until it transformed into gel. The mechanical stirring is stopped, and the temperature is raised to 423 K. The temperature of the gel was raised for 180 min until a self-sustaining auto-combustion reaction was obtained. After the gel was burnt, a homogenous dried powder was obtained. The powder was ground by mortar for fine powder grains.

Prepared $\text{Ni}_{0.5}\text{Zn}_{0.5}\text{Fe}_2\text{O}_4$ nanocomposites were added to commercial polyester in acetone solution with 0.0, 10.0, 20.0, and 40.0 wt% percentages and mechanical stirring to ensure a homogeneous solution. At last, the solution is kept in the air for 24 h to obtain a dried polyester composite. Each prepared composition is divided into 5 different samples with cylindrical shapes fitted to the collimator used in gamma-ray shielding investigation.

2.2. Composite characterization techniques

The physiochemical structure and properties of Polyester doped nanofiller composite ($\text{Ni}_{0.5}\text{Zn}_{0.5}\text{Fe}_2\text{O}_4$) samples are first characterized by x-ray photoelectron spectroscopy analyses (XPS), x-ray diffraction (XRD), density measurements, Field emission scanning electron microscope (FE-SEM) and Transmission electron microscope (TEM). Second, gamma-ray shielding properties are investigated and the shielding parameters are determined.

XPS analysis is confirmed using a K-Alpha spectrometer with Thermo Fisher scientific type (Waltham, MA, USA) also it uses a monochromatic Al $K\alpha$ radiation source (200W). XRD data was obtained using a Smart Lab x-ray diffractometer with Philips PW 3710/31, Eindhoven type (Malvern Panalytical, Spectrics Group, Netherlands). The data is performed in the range $2\theta = 5^\circ$ to 75° at room temperature.

Samples densities were calculated using the Archimedes method also surface morphology of the composite samples was illustrated using a Philips XL 30 instrument with accelerating voltage 30 kV and magnification up to 400,000 x. TEM for polyester composite is carried out using JEOL, JEM-2100 with a 200 kV operating voltage.

XRD analysis data is utilized with the Scherer formula to calculate the crystal size of the nanofiller as the following equation:

$$t = \frac{n \lambda}{\beta \cos \theta} \quad (1)$$

Where λ is the x-ray wavelength, t is the crystal size (Cu $K\alpha$ radiation 0.154056 nm), β is the full-width half maximum of the diffraction peak at 2θ and n is the Scherer constant and is equal to 0.89 [43].

Polyester composite samples are investigated against Gamma-ray using both narrow beam transmission technique and sodium iodide detector in addition to two gamma-ray point sources Cs-137 (emits 662 keV) and Co-60 (emits 1173 and 1332 keV).

Narrow beam transmission technique is granted by a lead collimator. The collimator provides the narrow gamma-ray beam from the point source to the composite sample and it has a 0.5 cm diameter (open hole between point source and sample). The distance between the point source and detector L_1 is equal to 10 cm while the distance between the sample and detector L_2 is equal to 5 cm. To ensure a narrow beam is delivered to the detector the point source and composite sample are stacked in the lead collimator and the collimator hole lies above the detector crystal.

The Bicron Scintillation detector is well calibrated and based on 76×76 mm crystal also 0.6 cm copper thickness protecting against induced x-ray in addition, 5 cm lead bricks provide isolation between all instruments and environmental radiation to prevent affecting measurements. Tennelec high voltage power supply (model, TC 952) with HV digital display also Nuclear Enterprises main shaping amplifier (model NE-4658) is connected to the detector. The setup of the sodium iodide detector is connected to a Nuclease PCA-800 computer-based and 8192 multichannel analyzer also a color graphical display of spectra and high-level technical operation features and an Epson LX-80 printer as shown in figure 1.

The energy calibration of the detector is done using radioactive calibration sources starting with Cs-137 in channel 662 KeV followed by Co-57 in channel 122 KeV. The detector has a sensitivity calibration using four artificial standards prepared from a series of certified reference samples with certain U, Th, Ra, and K concentrations obtained from the International Atomic Energy Agency (IAEA). The sensitivity is obtained by

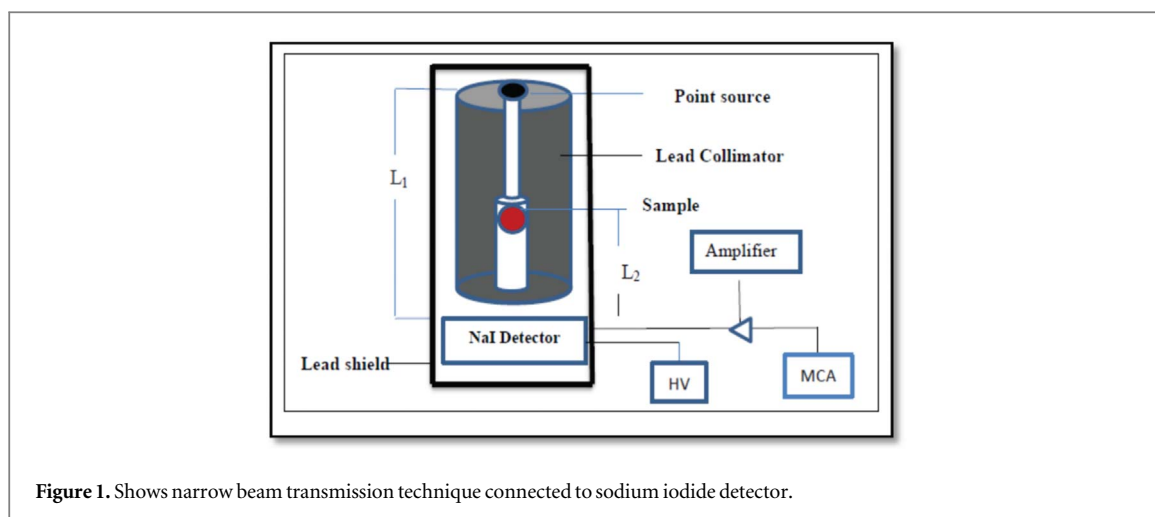


Figure 1. Shows narrow beam transmission technique connected to sodium iodide detector.

measuring four standards two times at 1000 s for each standard. The average gross count is recorded in a certain region of interest and normalized per unit mass. The expected measurement error is equal to 10% [44].

2.3. Shielding parameters calculation

The probability of absorbing or scattering collimated gamma-ray with and without polymer composite samples per unit length is described and calculated using Lambert–Beer equation (2) called linear attenuation coefficient (LAC) [45].

$$I = I_0 e^{-\mu X} \quad (2)$$

where I_0 is the collimated intensity without polyester sample, and I is the collimated intensity with polyester sample also X is the polyester sample thickness.

The measured densities of the polyester composite samples enable us to calculate a more precise and characterizing shielding parameter called mass attenuation coefficient (MAC). It can be calculated from equation (3).

$$\mu_m = \frac{\mu}{\rho} \quad (3)$$

Where μ is the linear attenuation coefficient and ρ is the density of the sample [46].

The ability or efficiency of an attenuator material to stop or decrease the intensity of gamma-ray to half or one-tenth of its original intensity is called Half value layer (HVL) and tenth value layer (TVL) respectively also it can be calculated from equations (4) and (5).

$$HVL = \frac{\ln(2)}{\mu} \quad (4)$$

$$TVL = \frac{\ln(10)}{\mu} \quad (5)$$

The interaction of gamma-ray with the attenuator material thickness determines the amount of absorbed or scattered radiation in that material moreover, transmitted radiation depends on those interactions in addition to the efficiency of gamma-ray shielding. These interactions can be described by the MFP parameter and can be calculated from equation (6). Where μ is the linear attenuation coefficient [47].

$$MFP = \frac{1}{\mu} \quad (6)$$

3. Results and discussion

3.1. Structure characterization

3.1.1. XPS analysis

XPS analysis gives a lot of information about the elemental composition of the studied material in addition to its distribution along the surface also it illustrates the chemical states of these elements as the binding energy of the core electron affects with chemical environment of the atom. XPS is extremely sensitive and efficient in filtering

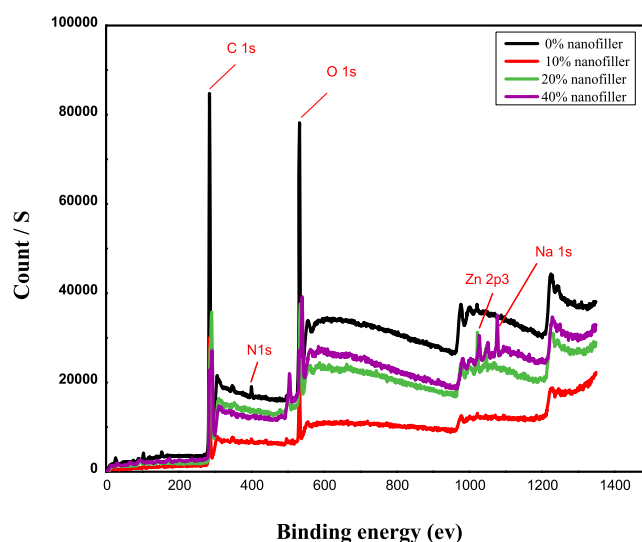


Figure 2. XPS for Polyester without and with different concentrations of $\text{Ni}_{0.5}\text{Zn}_{0.5}\text{Fe}_2\text{O}_4$ nanofiller.

Table 1. Atomic composition percent of polymer doped different concentrations of $\text{Ni}_{0.5}\text{Zn}_{0.5}\text{Fe}_2\text{O}_4$ nanofiller.

$\text{Ni}_{0.5}\text{Zn}_{0.5}\text{Fe}_2\text{O}_4\%$	0% (poly)	10%	20%	40%
Atomic composition %				
C	76.45	80.26	80.51	74.33
O	22.05	16.9	17.88	21.12
N	1.49	2.53	—	—
Zn	—	0.3	1.61	1.41
Na	—	—	—	3.13

out signals from the atoms of the analyzed sample [48]. Although XPS detects elements near the surface region, if the sample has a homogenous structure, it will determine the elemental composition of all sample structures.

The Atomic composition of polyester (polyethylene terephthalate) doped with different concentrations of $\text{Ni}_{0.5}\text{Zn}_{0.5}\text{Fe}_2\text{O}_4$ nanofiller is determined by XPS analysis in table 1 and figure 2.

XPS analysis pattern of the polymer doped nanofiller with different concentrations (0, 10, 20, and 40%) showed sharp peaks at 285.3, 285.21, 289.59, and 289.14 which corresponded to the characteristic peaks of C 1s in addition to 532.81, 532.41, 534.87 and 539.92 sharp peaks corresponding to the characteristic's peaks of O 1s. while two peaks 400.37 and 400.26 appear at 0% and 10% doping nanofiller corresponding to peaks of N 1s also 1022.82, 1023.25 and 1028.27 peaks appear at 10%, 20% and 30% doping which corresponding to peaks of Zn 2p3. At last, only one peak appears at 1077.96 which corresponds to the characteristic peak of Na 1s in the sample doped with 40% nanofiller. The peak positions have a negligible shift, and no other peaks are noticed in the pattern [49].

The presence of a Zinc atom and iron atom in the polyethylene terephthalate structure confirms the successful doping of $\text{Ni}_{0.5}\text{Zn}_{0.5}\text{Fe}_2\text{O}_4$ nanofiller. The C 1s, C 1s scan A, C 1s scan B, and C 1s scan C spectrums are shown in figures 3(a)–(d) which illustrate the deconvolution of the spectrum into four Gaussian peaks. The C 1s peak appeared at 284.22 eV due to the polyester structure (0% nanofiller doping) which refers to C=C chemical bonds (figure 3(a)) [50]. The peaks that appeared at 284.35, 288.23, and 292.33 eV are due to different doping of nanofiller (10, 20, and 40%) (figure 3(a)). The peak positions at 284.2 eV and 284.3 for 0% and 10% doping nanofillers are almost the same, but the intensity decreased due to doping. At 20% doping, the peak position shifted from 284.2 to 288.23 eV. Also, at 40% doping, the peak position shifted from 284.2 to 292.33 eV. The more nanofiller doping, the more shifted peaks, and less intensity are obtained.

Similarly, at C 1s scan A and C 1s scan B have a peak position of 285.85 eV and 288.34 eV for polyester structure (0% nanofiller) which indicate C-O and O-C=O chemical bond respectively [51–54], the peaks position of binding energy for both 0% and 10% nanofiller doping are considered almost the same (285.85 eV and 285.37) but there is a decrease in the intensity of the peak also doping with 20% and 40% nanofiller shift the peaks. At 20% C 1s scan A, the peak shifted towards increasing and equal to 290.34 eV while at the same doping the peak shifted towards decreasing and equal to 284.68 eV in C 1s scan B. Similarly, at 40% doping, the peaks

(a)

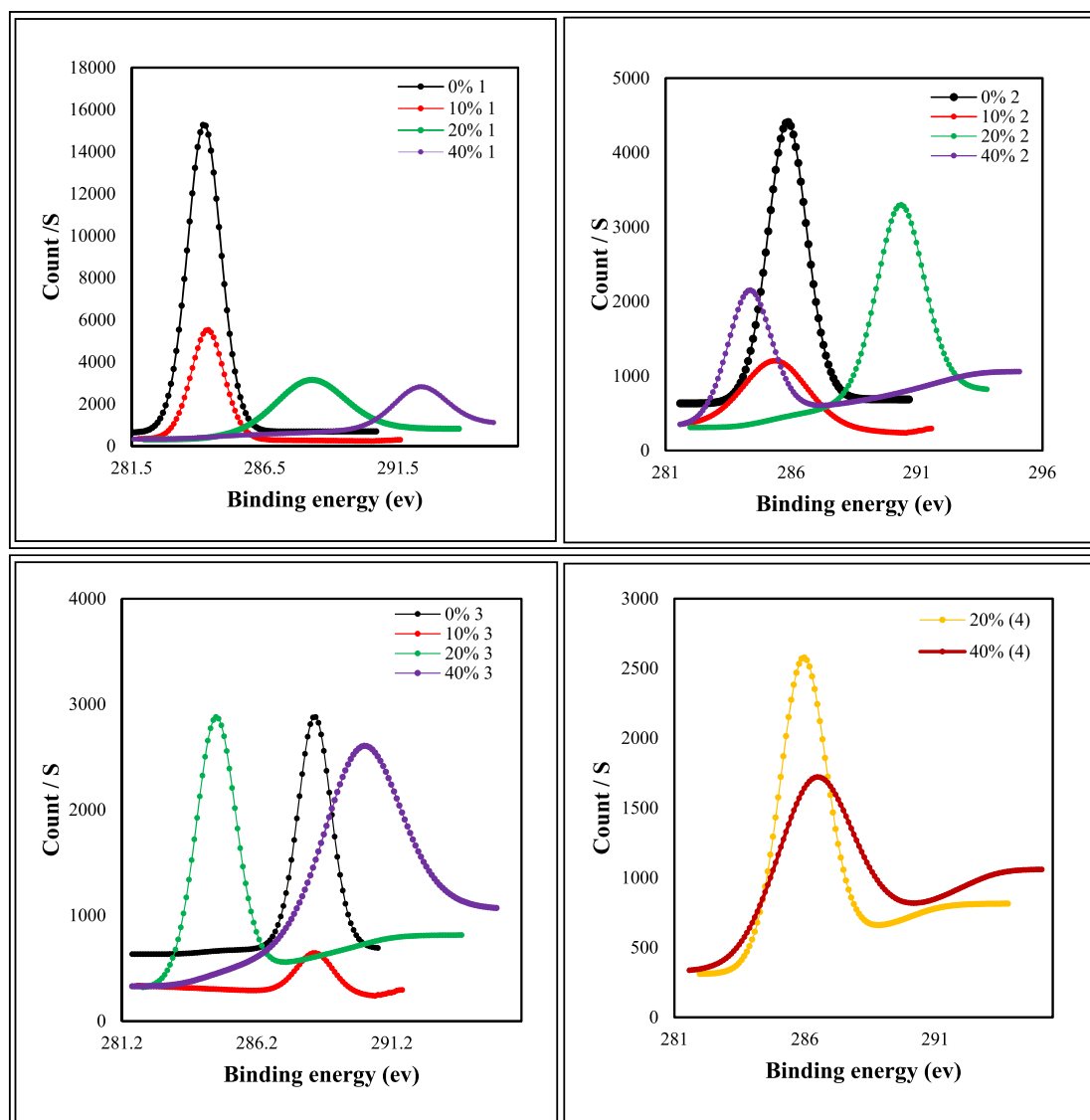


Figure 3. (a) XPS analysis of polyester doped different concentrations of nanoparticles for Carbon element where 1, 2, 3, and 4 numbers are C1s, C1s scan A, C1s Scan B, and C1s scan C. (b) XPS analysis of polyester doped different concentrations of nanofiller for Nitrogen element. (c) XPS analysis of polyester doped different concentrations of nanofiller for oxygen element where 1, 2, and 3 numbers are O1s, O1s Scan A, and O1s Scan B. (d) XPS analysis of polyester doped different concentrations of nanofiller for Zinc element where 1, 2, 3, and 4 numbers are Zn 2p, Zn 2p scan A and Zn 2pScan B. (e) XPS analysis of polyester doped different concentrations of nanofiller for iron element (Fe) where 1, 2, 3, 4, 5, and 6 numbers are Fe2p, Fe 2p scan A, Fe 2pScanB, Fe 2p scan C, Fe 2p scan D and Fe 2pscan E.

shifted towards decreasing at C 1s scan A and equal to 284.33 ev while at the same doping in C 1s scan B, the peak shifted towards increasing and equal to 290.12 ev.

At last, two peaks appeared at binding energy 285.94 and 286.42 ev at 20% and 40% nanofiller doping which indicates C=N and C-O chemical bonds respectively (figure 3(b)) [55, 56].

The O1s peak positions of binding energy were found between 529 ev and 535 ev [57] as shown in figure 3(c). The deconvolution of O1s discovers three different types of O atoms with binding energy 531.65 ev, 533.05 ev, and 536.32 ev. The last one appears only at high doping (40%) of nanofiller, and it may be due to oxygen transferred to the structure from nanofiller doping. The peak position at 531.65 ev is attributed to the O-H bonds [58, 59]. The peak is shifted in the direction of growing with excess nanofiller doping as follows: 531.65 at 0% doping, 531.42 ev at 10% doping, 532.56 ev at 20%, and 539.02 ev at 40%. Also, the intensity of the peaks decreased with the increase of nanofiller doping.

Similarly, the peak position at 533.05 ev is attributed to the C-O-C bond which indicates the existence of O and C [60]. The peak shifted as a result of doping nanofiller to polyester and the intensity of the peaks decreased. It shifted to 532.83 ev at 10%, 535.2 ev at 20% and 532.05 at 40% doping.

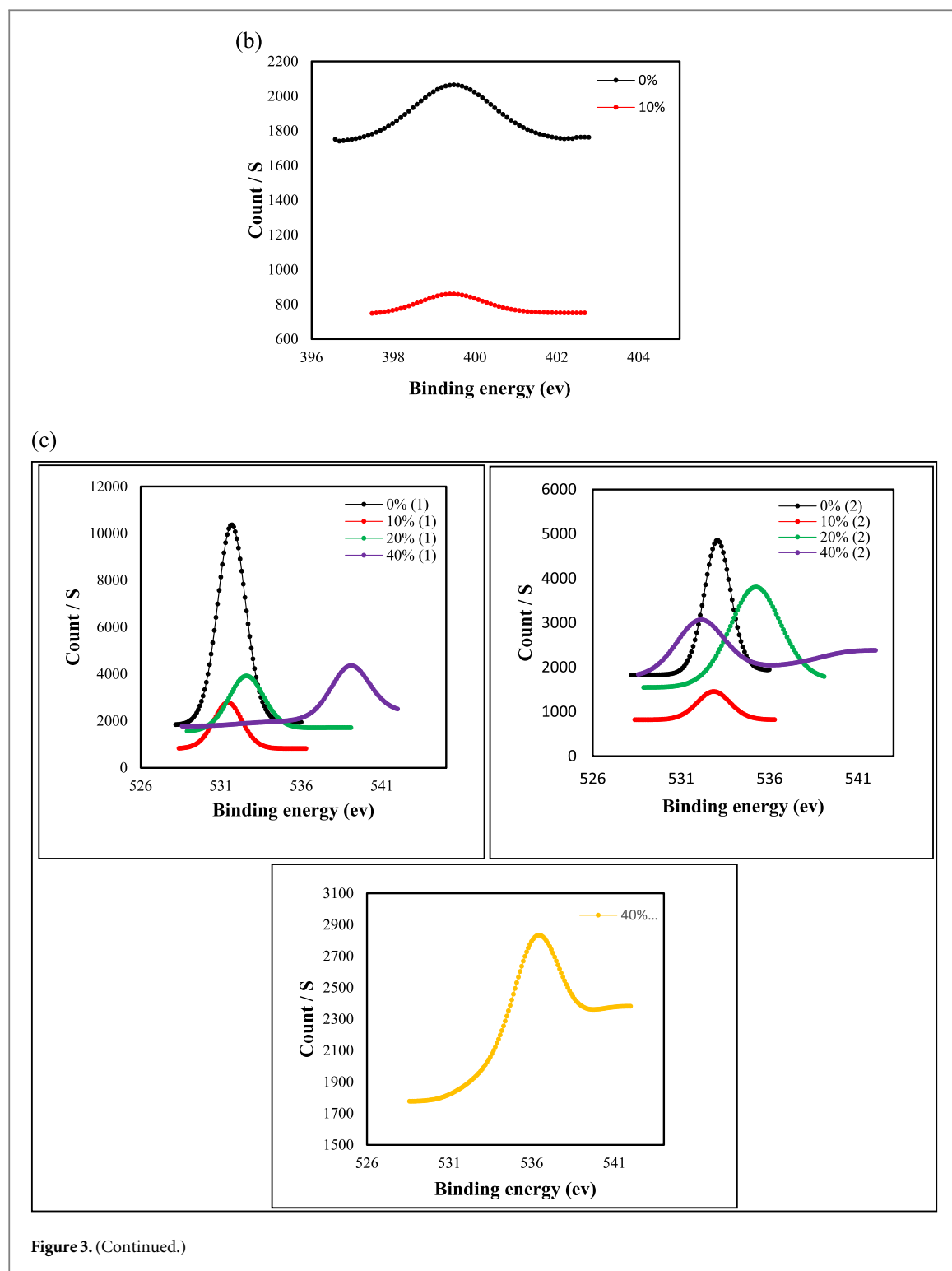


Figure 3. (Continued.)

As illustrated in figure 3(d), the photoelectron peak of Zn 2p has a splitting to Zn 2p_{3/2} and Zn 2p_{1/2} which was observed at binding energy 1021.7 eV and 1043.77 eV respectively, for 10% doping nanofiller [61]. Zn 2p_{3/2} photoelectron has a shift due to excess nanofiller doping (1022.57 eV at 20% and 1027.31 eV at 40%) also Zn 2p_{1/2} photoelectron has a shift to 1045.44 eV at 20% and 1050.11 eV at 40% nanofiller doping [62]. Another two photoelectrons Zn 2p_{3/2} and Zn 2p_{1/2} appear for 40% nanofiller doping at 1022.58 eV and 1045.06 eV That means the chemical environment doesn't change. This result is in agreement with the reported data [63].

High-resolution spectrum of Fe 2p deconvoluted to six peaks at binding energies 709.18, 721.67, 724.23, 712.63, and 718.95 eV including satellite peak (727.97 eV) which shifted to 731.3 eV at 20% doping and 712.26 eV at 40% doping. The peaks at 724.23 eV and 712.63 eV (10% doping) are due to the splitting of Fe 2p into Fe 2p_{1/2} and Fe 2p_{3/2} respectively, that ensure the existence of Fe³⁺ and Fe²⁺ ions in the polyester nanofiller composite. Fe 2p_{1/2} peak has a shift to 712.2 for 20% doping and 724.88 eV for 40% doping also Fe 2p_{3/2} has a shift to 716.15 eV for 20%

(d)

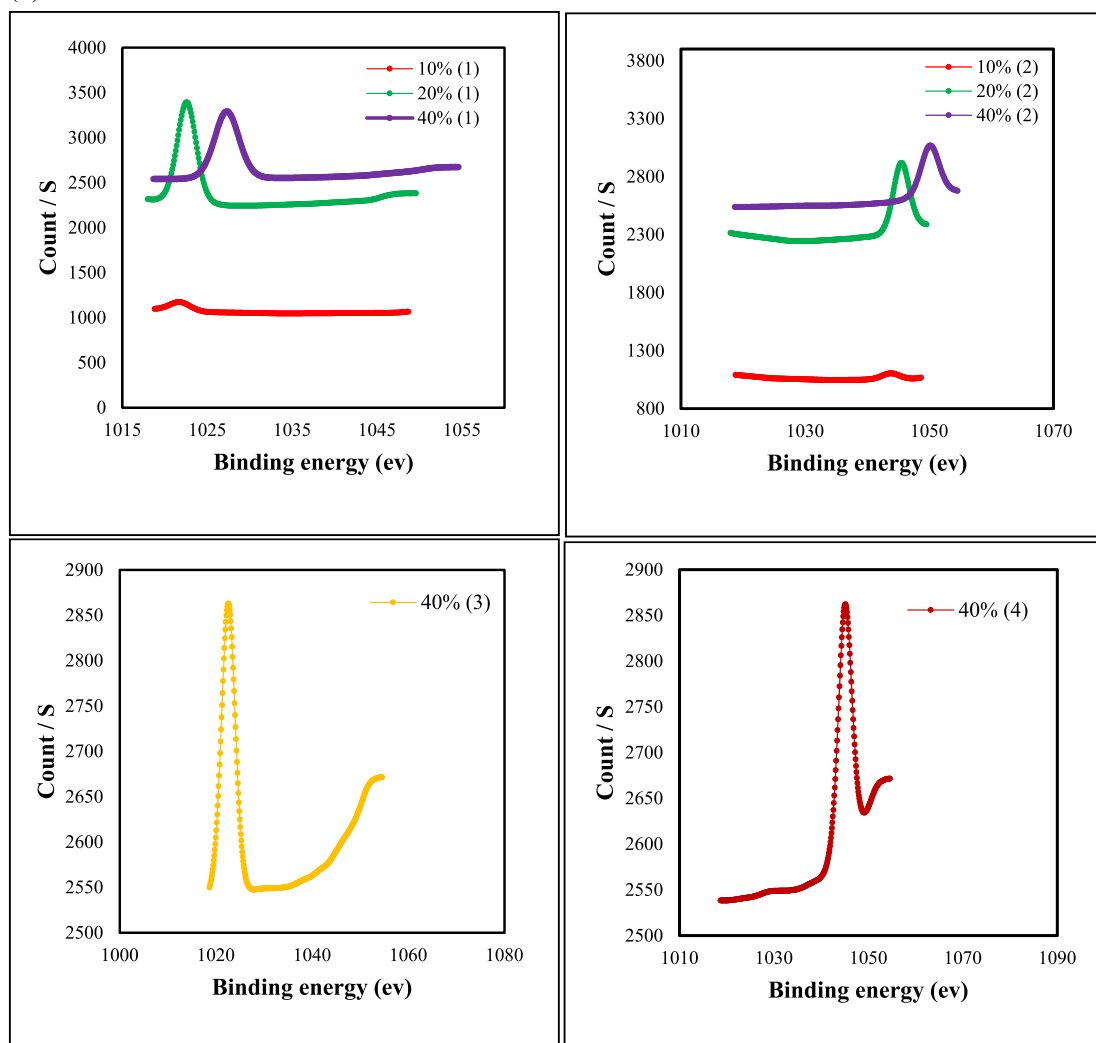


Figure 3. (Continued.)

doping and 720.96 eV for 40% doping [64]. The peak at 718.95 is attributed to Fe^0 and shifted to 721.84 eV for 20% doping and also shifted to 716.73 eV for 40% doping. The peak at 709.18 is attributed to $\text{Fe}2p_{3/2}$ and shifted to 731.3 eV at 20% doping also shifted at 712.26 eV at 40% doping [65] while the peak at 721.67 is attributed to $\text{Fe}2p_{1/2}$ for 10% doping and shifted to 709.3 for 20% doping also shifted at 729.38 eV for 40% doping [66, 67] figure 3(e).

3.1.2. X-ray diffraction analysis

The crystallographic structure of the polyester/nanofiller composite is investigated using XRD analysis in addition to physical and chemical properties [68]. Figure 4 shows the XRD analysis of the polyester composite. XRD analysis proves the successful doping of $\text{Ni}_{0.5}\text{Zn}_{0.5}\text{Fe}_2\text{O}_4$ in the polyester structure as the major peaks of nanofiller appear in the XRD graph and this result matches with XPS analysis. The main peaks appeared at $2\theta = 35.29^\circ$ (311) and $2\theta = 62.28^\circ$ (440) also the other peaks appeared at $2\theta = 29.95^\circ$ (220), 36.90° (222), 42.89° (400), 53.53° (422) and 56.72° (511) in addition to two peaks corresponding to polyethylene terephthalate polyester (19.35° and 25.05°). The intensities of XRD peaks increase with increasing nanofiller doping while polyester peaks decrease, and this illustrates the dominant rule of nanofiller with a high concentration in the polyester structure. Based on XRD data, the crystal size of the nanofiller was determined by employing the Scherrer formula (1). The average crystal size of the nanofiller is equal to 26 nm.

3.1.3. Density measurements

Archimedes method is utilized for measuring the densities of the polyester composite samples in which two different liquids water and xylene are used. The data are graphed as shown in figure 5. The graph demonstrated that the density of the samples increases when the percentage of nanofiller increases in the polyester composite.

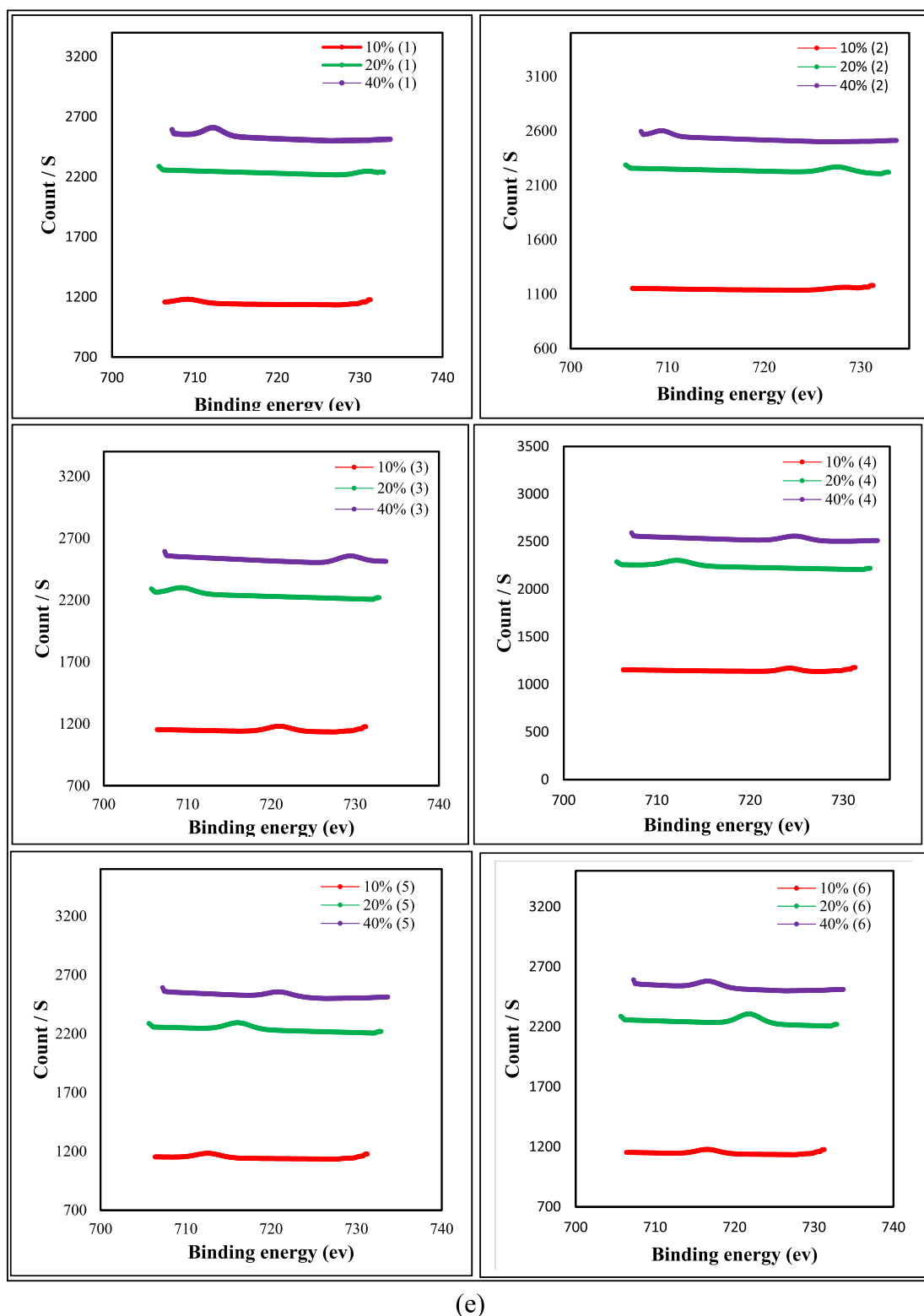


Figure 3. (Continued.)

It increases from 1.247 g/cm³ for only polyester without nanofiller to 2.013 g/cm³ for polyester doped 40% nanofiller.

3.1.4. Morphology investigation

Figure 6 shows the morphological investigation of polyester nanofiller composite samples using both Transmission and scanning electron microscope. SEM images for 0%, 20%, and 40% nanofiller. It showed that nanofiller propagates homogenously in the polyester structure and there are leaves phase growth with increasing doping of nanofiller figure 6(b). At a high concentration (40%) of nanofiller, the leaves phase fills all structures,

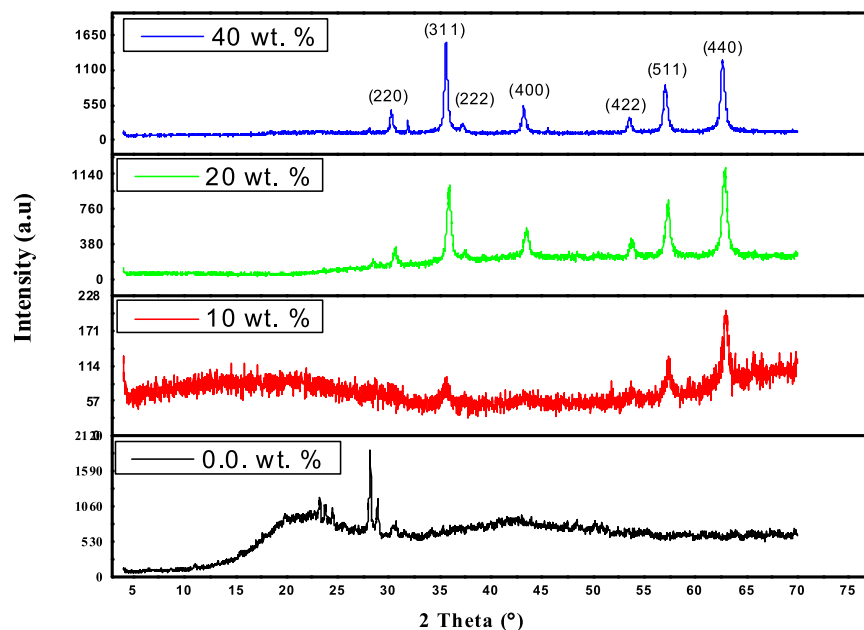


Figure 4. XRD analysis for polyester with and without different concentrations from $\text{Ni}_{0.5}\text{Zn}_{0.5}\text{Fe}_2\text{O}_4$ nanofiller.

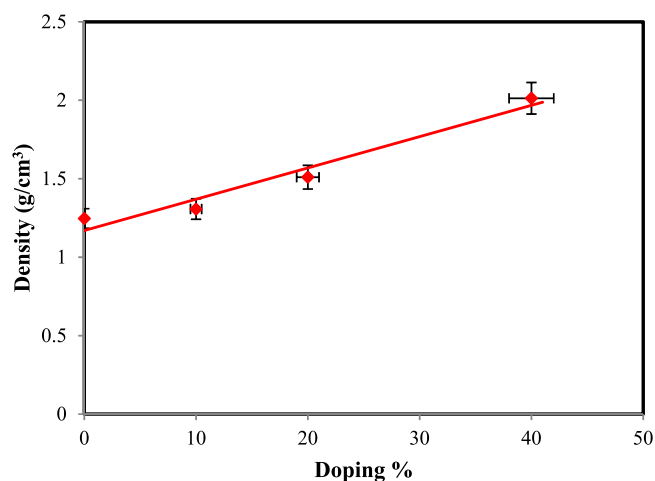


Figure 5. Polyester doped different concentration from nanofiller ($\text{Ni}_{0.5}\text{Zn}_{0.5}\text{Fe}_2\text{O}_4$).

and that ensures the high interaction and corresponding changes of the polyester structure due to doping with nanofiller figure 6(c). Figure 6(d) showed TEM for the polyester nanofiller composite, and it demonstrated the appearance of dark spots which represent the agglomeration of $\text{Ni}_{0.5}\text{Zn}_{0.5}\text{Fe}_2\text{O}_4$ nanofiller in the polyester structure and form large flake-like micrometer-sized particles.

3.2. Shielding calculation of polyester/nanofiller composite

Polyester/nanofiller composite is tested as a gamma-ray shielding material as an advanced application. Shielding parameters for the composite are determined at 662 keV, 1173 keV, and 1332 keV. Based on Lambert–Beer equation (2), the relation between Relative intensity (I/I_0) (where I count with sample thickness and I_0 count without sample thickness) and sample thickness (cm) are plotted, and the exponential slope is equal to linear attenuation coefficient (LAC) (See figure 7). The relative intensity of the gamma-ray decreased as the thickness of the composite sample thickness increased. On the contrary, relative intensity increases as the energy of gamma-ray increases. LAC represents the probability of absorption or scattering of gamma-ray per unit length from polyester composite samples.

LAC (cm^{-1}) of the polyester doped different concentrations of nanofiller (0.0%, 10.0%, 20.0%, and 40.0%) for each gamma-ray energy (662 KeV, 1173 KeV, and 1332 KeV) estimated from exponential relation between Relative intensity (I/I_0) and sample thicknesses (X (cm)) are graphed in figure 8 and shown in table 2. LAC of the

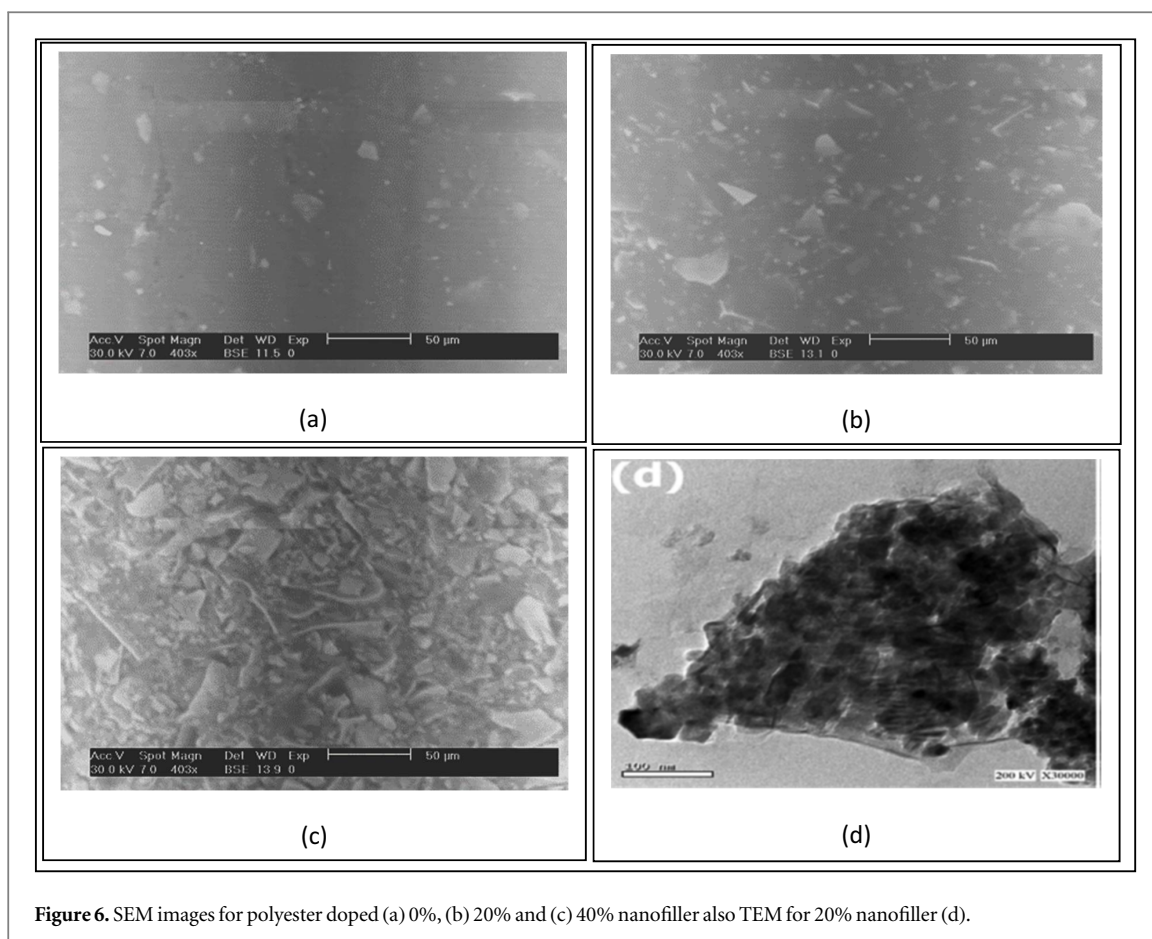


Figure 6. SEM images for polyester doped (a) 0%, (b) 20% and (c) 40% nanofiller also TEM for 20% nanofiller (d).

composite was increased for each energy due to the increasing doping of the nanofiller in the polyester structure also LAC decreased as energy increased for all doping concentrations as LAC depend on incident photon energy, atomic number, and density of the material. The last two factors are increased by increasing doping with nanofiller (see figures 5 and 6). The remarkable increase in LAC value was the sample with the highest nanofiller doping (40%). LAC increases from 0.0864 cm^{-1} for only polyester (0%) to 0.1448 cm^{-1} for polyester-doped 40% nanofiller at 662 KeV. Similarly, at 1173 KeV, LAC increased from 0.0661 cm^{-1} for polyester to 0.1089 cm^{-1} for 40% doping also at 1332 KeV, LAC increased from 0.0627 cm^{-1} for polyester to 0.1042 cm^{-1} for 40% doping. The outcome result described that adding nanofiller with a high percentage in polyester structure enhances linear attenuation coefficient in addition to figure (6) described the successful doping of nanofiller in the polyester structure.

Since LAC depends on the density of the absorber material, it can be expressed in another factor called mass attenuation coefficient (MAC) and can be given experimentally by using equation (3) from the determined values of LAC. Table 3 shows experimental MAC values in cm^2/g for each doping percentage and different energies. The relation between nanofiller percentages in the polyester composite and MAC is drawn in figure 9. The figure shows that MAC values increase with the increase of nanofiller doping in the polyester structure. it follows the same behavior of LAC and the same cause of increasing which is the excess doping of nanofiller in the structure and the increase of density as shown in figure 10. MAC increases from $0.0692 \text{ cm}^2/\text{g}$ for polyester only to $0.0719 \text{ cm}^2/\text{g}$ for 40% nanofiller doping at 662 keV. Similarly, it increases from $0.053 \text{ cm}^2/\text{g}$ for polyester to $0.0541 \text{ cm}^2/\text{g}$ for 40% doping at 1173 keV. Also, it increases from $0.0502 \text{ cm}^2/\text{g}$ for polyester to $0.0517 \text{ cm}^2/\text{g}$ for 40% doping at 1332 keV. The error in MACs values does not exceed 2%. The errors come from the measured density, thickness, count without thickness, and count with thickness. Archimedes principle used in the density measurements has an accuracy of 0.001 while measured thickness has an accuracy of 0.002.

Verification must be introduced even in experimental-based results. The elemental fractions of polyester/nanofiller composite specified by XPS analysis for different nanofiller doping are utilized in calculating MAC theoretically. Theoretical MAC was calculated using two programs for more verification (Physics-X and X-com programs). XCOM program was first published by Berger and Hubbell (1987, 1999) [69, 70] for calculating photon cross sections for the photoelectric effect, scattering, and pair production. It calculates the total attenuation in the range from 1 keV to 100 GeV for elements, compounds, and mixture of elements while the

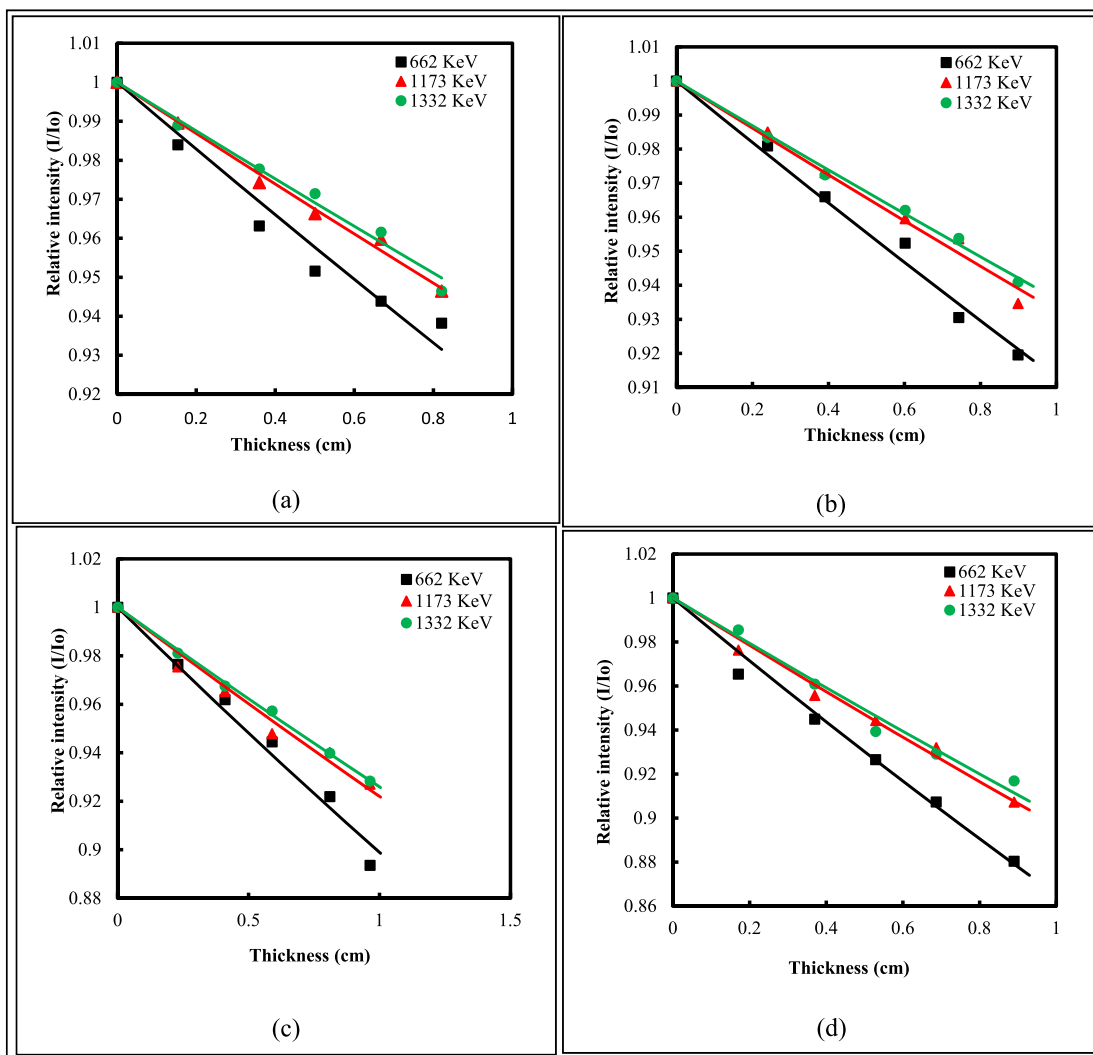


Figure 7. Gamma transmission through polyester doped different concentrations of nanofiller percent at different thicknesses and different energy where (a) 0% nanofiller, (b) 10% nanofiller, (c) 20% nanofiller, and (d) 40% nanofiller.

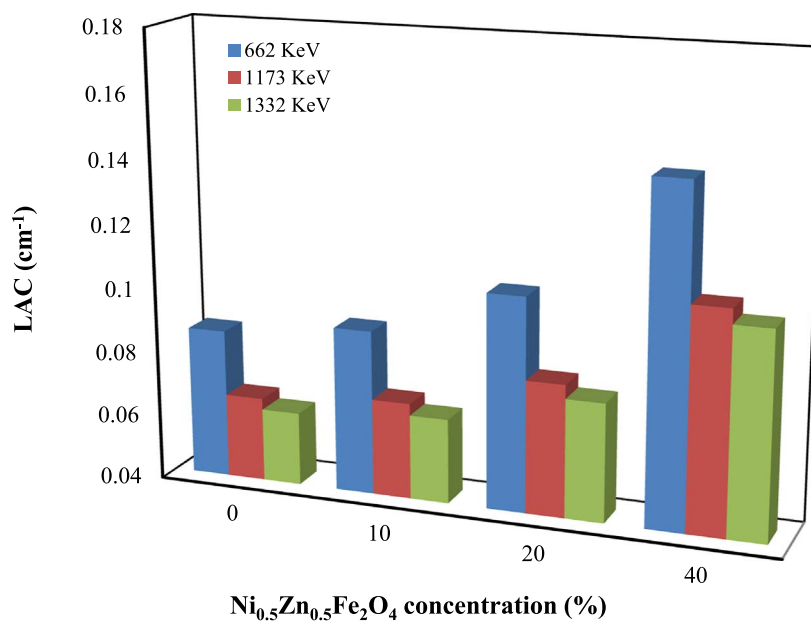


Figure 8. LAC (cm^{-1}) for polyester doped different concentrations of nanofiller at different energy.

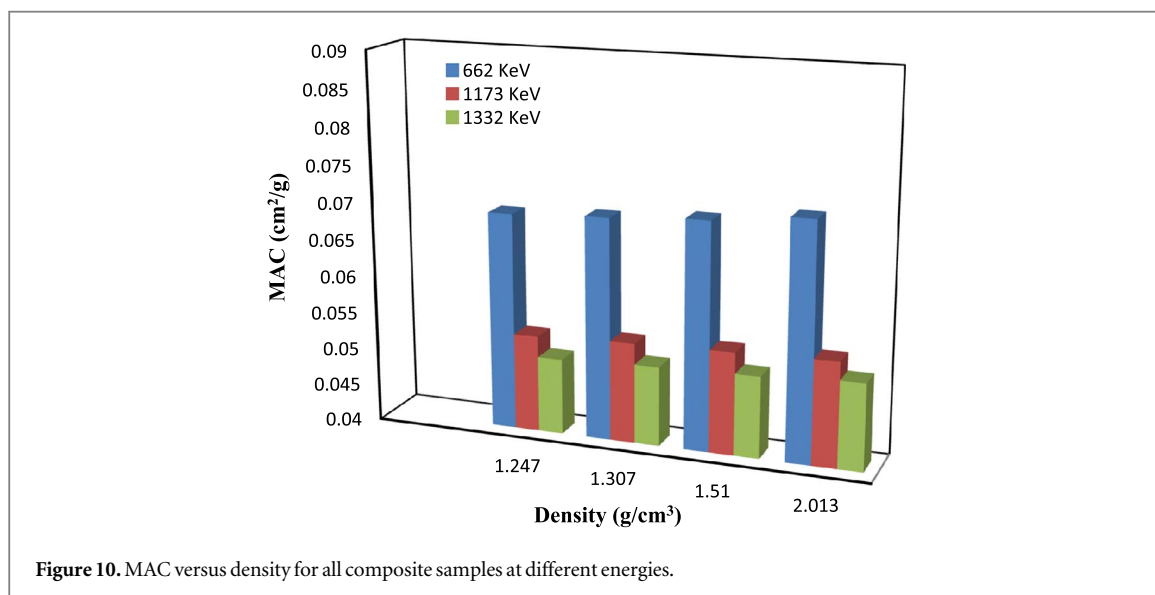
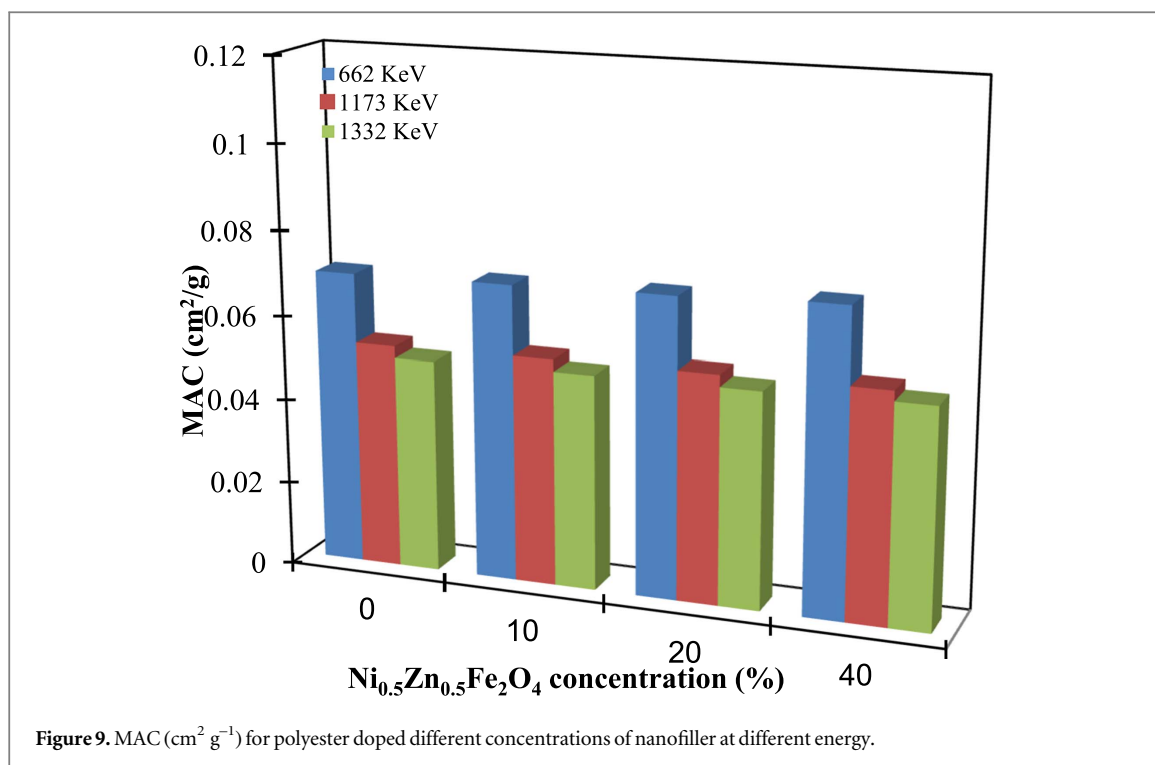


Table 2. LAC for polyester/composite structure at different doping and different energies.

Polyester/nanofiller composite	Linear attenuation coefficients (cm^{-1})		
	662 KeV	1173 KeV	1332 KeV
0% doping	0.0864 ± 0.0005	0.0661 ± 0.0009	0.0627 ± 0.0010
10% doping	0.0912 ± 0.0005	0.0699 ± 0.0009	0.0663 ± 0.0010
20% doping	0.1066 ± 0.0006	0.0812 ± 0.0010	0.0769 ± 0.0010
40% doping	0.1448 ± 0.0007	0.1089 ± 0.0010	0.1042 ± 0.0020

physics-X program is a computer software program calculating Photon Shielding and Dosimetry (PSD) for any compound or composite [71]. All shielding parameters can be obtained from the Physics X program.

The experimental and theoretical MAC was compared to each other at 662 KeV as shown in figure 11 also the deviation between theoretical and experimental values is listed in table 3. Figure 11 shows that theoretical and experimental MAC increases when the percentage of nanofiller doping increases in the polyester structure. Both

Table 3. Experimental and theoretical MAC (cm²/g) for all polyester samples doped with different concentrations of nanofiller at different energies (the errors in MACs values do not exceed 2%).

composite MAC (cm ⁻¹)	Polyester (0%)			10% doping			20% doping			40% doping		
	662 (keV)	1173 (keV)	1332 (keV)	662 (keV)	1173 (keV)	1332 (keV)	662 (keV)	1173 (keV)	1332 (keV)	662 (keV)	1173 (keV)	1332 (keV)
Experimental	0.0692	0.0530	0.0502	0.0698	0.0535	0.0507	0.0706	0.0538	0.0510	0.0719	0.0541	0.0517
XCOM	0.0772	0.0588	0.0551	0.0772	0.0588	0.0551	0.0771	0.0587	0.0550	0.0770	0.0586	0.0549
Phy-X	0.0774	0.0593	0.0557	0.0773	0.0592	0.0556	0.0771	0.0590	0.0554	0.0769	0.0590	0.0554
Deviation (%)	11.5	10.1	9.7	10.5	9.9	8.6	9.2	9.1	7.8	7.1	8.3	6.1
Exp and xcom												
Deviation (%)	11.8	11.9	10.9	10.7	10.6	9.7	9.2	9.7	8.5	6.9	9.1	7.1
Exp and Phy-X												

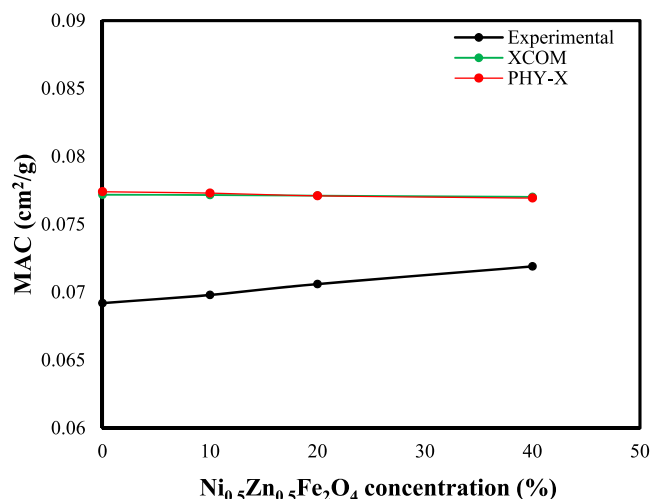


Figure 11. Comparison between theoretical and experimental MAC values at different nanofiller concentrations (%).

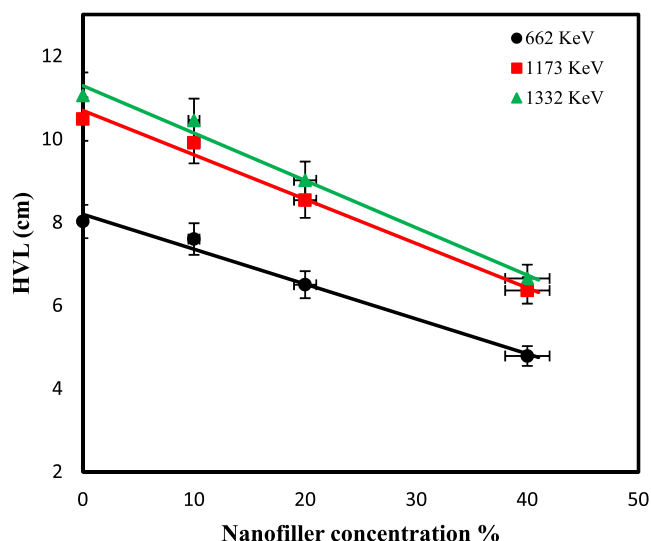


Figure 12. HVL values for polyester doped different concentrations of nanofiller up to 40% at different energies.

theoretical values obtained from XCOM and Physics-X programs are almost the same while there is a little difference between them and experimental data. The deviation does not exceed 12% for both XCOM and Physics-X (see table 3). This little difference is due to atomic composition determined from XPS analysis that utilized in theoretical calculation from both programs illustrate the absence of Ni signals from XPS analysis although it exists in the composition of nanofiller doped in the polyester structure in addition to Fe signals does not exist in the overall atomic composition of samples although it exists in deconvolution of peaks for every element. Moreover, the existence of some elements in the structure does not participate in the ingredients of the preparation method like Na which may be contaminated in the commercial polymer.

At last, the experimental result is closer to reality (more accurate) than the theoretical value (reliable result). The absence of Ni in the overall atomic composition and the appearance of a small amount of Na signals in the sample structure produce errors in the theoretical MAC measurements.

The deviation between both theoretical values (XCOM and Physics X) and experimental values is almost the same while the deviation between both theoretical values does not exceed 1%.

The effectiveness of polyester doped different percentages of nanofiller as gamma-ray shield is confirmed by two important parameters Halfvalue layer (HVL) and tenth value layer (TVL). These two parameters are extremely used quantitative and characterize factors for investigating the penetration capability of gamma-ray through a specific material. When gamma-ray with specific intensity incident on absorbing material, it attenuated using this material thickness by 50% and 10% from its original value and it is referred to as HVL and TVL respectively.

The previously calculated LAC and equations (4), and (5) are utilized in determining HVL and TVL factors. As shown in figures 12 and 13, HVL and TVL variation with both nanofiller concentration in the polyester

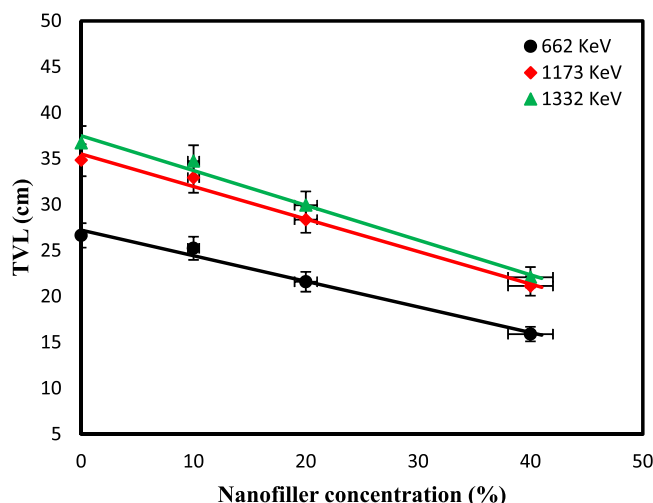


Figure 13. TVL values for polyester doped different concentrations of nanofiller up to 40% at different energies.

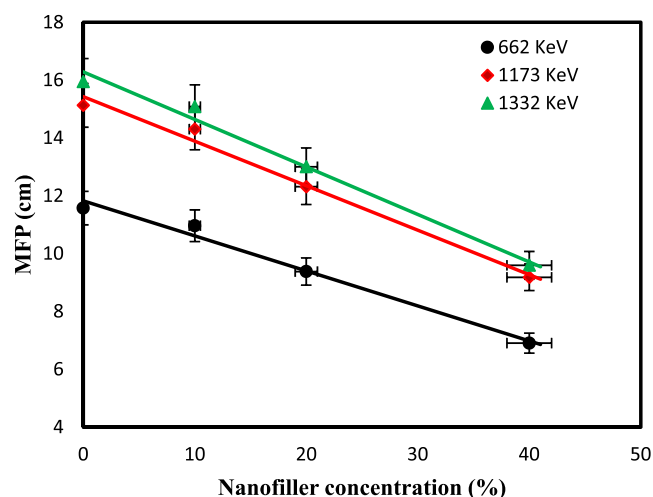


Figure 14. MFP values for polyester doped different concentrations of nanofiller up to 40% at different energies.

structure and energy. The accretion of nanofiller in the structure lowers or decreases HVL and TVL values and this increases the efficiency of polyester structure against gamma-ray. This result is due to the increasing density of the samples and doping with nanoscale material [72]. Moreover, HVL and TVL values were reduced with excess energy. HVL values reduced from 8.06 cm for polyester to 4.78 cm for 40% nanofiller doping also TVL reduced from 26.6 cm for polyester to 15.9 cm for 40% nanofiller doping at 662 keV. Eventually, one more parameter which is called mean free path and defined as the distance between two successful interactions is calculated using equation (6). As illustrated in figure 14 MFP displays the same attitude as HVL and TVL as it decreased with excess nanofiller doping and raised as energy increased. It reduced from 11.5 cm for polyester to 6.9 cm for 40% nanofiller doping at 662 keV.

4. Conclusion

A lightweight, low-cost, sustainable, and large-scale application polyethylene terephthalate / $\text{Ni}_{0.5}\text{Zn}_{0.5}\text{Fe}_2\text{O}_4$ gamma-ray shield is constructed. Nano-size $\text{Ni}_{0.5}\text{Zn}_{0.5}\text{Fe}_2\text{O}_4$ filler is guaranteed by the sol-gel auto-combustion method (26 nm). The successful doping of nanofiller to the polyester structure is ensured by x-ray photoelectron spectroscopy analysis as Zn peaks appeared in the overall atomic composition while Fe signals appeared in the deconvolution of the peaks. The composite showed good shielding properties which increased with the increasing of nanofiller content. So the sample with 40% nanofiller doping is the best as it has higher density, higher Mass attenuation coefficient, lower half value layer, Tenth value layer, and Mean free path. The

comparison between the experimental Mass attenuation coefficient and the theoretical value obtained from both XCOM and Physics-X programs showed a small difference between them reaching 12%. These results are from the absence of Ni in the overall atomic composition and the appearance of a small amount of Na signals in the sample structure. The appearance of Na signals might be due to commercial polyester contamination.

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Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

ORCID iDs

Fawzy Hammad Sallam  <https://orcid.org/0000-0002-1510-5492>

Atef El-Taher  <https://orcid.org/0000-0003-0188-420X>

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